EEE4118F 2026
Controller Implementation and GA Report

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Contents

Chapter 1

Introduction

1.1 Background

The DC servo motor plant consists of various hardware subsystems, including operational-amplifier circuits, power amplifiers, and a tachometer, each possessing unique physical characteristics. At its core, the system relies on the conversion of electrical energy into mechanical torque, which is then used to overcome rotational inertia and damping to achieve a speed which is integrated to reach a target position.

The Robust Digital Position and Control project is structured into three distinct phases [1]:

1. **Modelling and system identification:** This phase involved modeling the system's open-loop parameters across four plant configurations (LD, LB, HB, and HD) to identify varying gains (A) and time constants (T) [2]. This work has been documented by Group 20.

2. **Robust control design:** This stage focuses on the theoretical design and simulation of a nested cascade architecture, comprising an inner-loop speed controller (G_i) and an outer-loop position controller (G_o) [3].

3. **Automation interface:** The final phase, which is outside the scope of this report, involves the introduction of automation via PLC and Human-Machine Interface (HMI) [1].

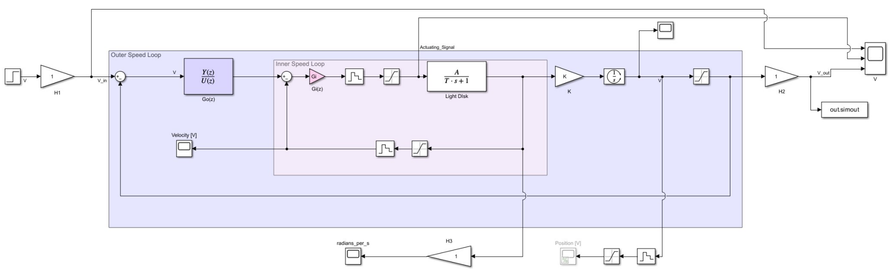


Figure 1.1: Open Loop System Model for the Servo Plant.

Figure 1.1 represents the cascade-controlled closed-loop model of the DC servo system, illustrating the parameters to be determined during the initial two phases of the project.

1.2 Aim

The aim of this report is to refine and implement a robust cascade control system for the speed and position regulation of a bidirectional DC servo motor across a variety of load configurations.

Chapter 2

Modelling and System Identification

This section summarises the process taken to identify the factors of the servo motor system's open loop system which include the plant transfer function, integrator gain, and sensor constants as detailed in the System Identification Report which was compiled by Group 20 [2].

2.1 Plant Identification

The DC servo system was modelled as a first-order Linear Time-Invariant (LTI) system, $P(s) = \frac{A}{1+Ts}$, across four distinct physical configurations: Light Disk (LD), Light Brake (LB), Heavy Brake (HB), and Heavy Disk (HD).

To establish a valid LTI operating range, a Ramp Test was performed. This test aided in identifying five operational regions, including a central dead-band between -1.7V and $+1.8$V where developed electromagnetic torque is insufficient to overcome static friction, resulting in the motor not spinning.

To determine the steady-state gain (A) and time constant (T) for each plant configuration, a Linear Least-Squares regression approach was used as it is statistically accurate compared to a manual graphical analysis.

As summarised in Table 2.1, each plant configuration was accurately modelled and validated by comparing the experimental step response data against the identified first-order LTI simulation.

Table 2.1: Identified System Parameters for Various Plant Configurations

System	**Steady State Voltage [V]**	A	**T[s]**
LD	2.4940	4.1567	1.3000
LB	2.4200	2.4200	0.2900
HB	1.7200	1.1466	0.1904
HD	4.8462	4.7327	2.1350

2.2 Sensor Constants (H_1, H_2, H_3)

Three sensor transfer functions were identified to characterise the signal conversions within the plant. H_1 represents the relationship between the input angle and voltage, H_2 represents the relationship between the output voltage and angle, and H_3 relates the speed voltage to the angular velocity. To minimise measurement error and ensure a statistically robust estimate, three measurements were taken at various operating points for each sensor, with linear regression applied to the resulting datasets. The sensor constants were determined to be:

$$H_1 = 0.0972 \text{ V}/^\circ, \qquad H_2 = 17.4204 \text{ }^\circ/\text{V}, \qquad H_3 = 42.3217 \ \frac{\text{rad/s}}{\text{V}}$$

2.3 Integrator Gain (K)

The integrator gain K relates the steady-state velocity voltage to the slope of the integrated position signal. To determine this value, a unit step input was applied to the system, and the gain was calculated using the relationship:

$$K = \frac{m}{V_{steady}}$$

where m is the slope of the position ramp and V_{steady} is the steady-state velocity voltage. To ensure accuracy and mitigate the effects of sensor noise, the slope m was estimated using linear least-squares regression (`polyfit`) over three distinct time intervals of the same response. The mean of these three independent estimates was then taken as the final value for K:

$$K = 7.1296$$

Chapter 3

Robust Controller Design

A cascade control architecture was adopted because, unlike a generic single-loop configuration, it increases the overall degrees of freedom available for tuning the system. This approach allows the inner-loop controller (G_i) to proactively reject disturbances and mitigate the effects of phase lag inherent in the system's mechanical dynamics such as damping, friction, and inertial loading before they impact the primary output of the system. By stabilising the inner loop first, the system's robustness to plant uncertainty due to the various physical configurations is significantly enhanced. Subsequently, the outer-loop controller is used for loop shaping, ensuring the overall system behavior satisfies the client's stringent performance requirements and frequency-domain design specifications.

3.1 Specifications

A Quantitative Feedback Method (QFT) was employed to design the cascade controller for the servo motor system. Tables 3.1 and 3.2 below, detail the time domain specificatons and frequency domain specifications which are to be met.

Table 3.1: Time Domain Specifications

ID	Parameter	Client Requirements
R1	Speed controller response time	1.5 s
R2	Overshoot in velocity loop	25 %
R3	Margin for digital design	15°
R4	Unsaturated step size	10°
R5	Overshoot in position loop	20%
R6	Position controller response time	1.0 s

Table 3.2: Frequency Domain Specifications

ID	Parameter	Frequancy Domain Specification
S1	Disturbance Rejection	$\left\lvert \frac{1}{1+L_\omega(s)} \right\rvert \leq -20$ dB, $\omega = 0.5$ rad/s
S2	Robustness	$\left\lvert \frac{1}{1+L_\omega(s)} \right\rvert \leq 3$ dB, $\forall\omega$

3.2 Controller Design Methodology

This section summarises the systematic process used to design the robust cascade controller. By partitioning the design into two distinct stages, the overall degrees of freedom are increased, allowing for independent optimization of speed and position characteristics.

3.2.1 Step 1: Inner-Loop Speed Controller, G_i

The inner-loop controller, G_i, is designed as a proportional controller to provide the system with a significantly higher bandwidth compared to the outer loop. This frequency separation ensures that the inner loop responds rapidly to disturbances before they can impact the position output.

A nominal plant was selected from the plant template, and the Quantitative Feedback Theory (QFT) approach was employed to satisfy specification S1 afrom Table 3.2. The performance was verified through step tests using the closed-loop speed transfer function:

$$T_i(s) = \frac{G_i(s)P(s)}{1 + G_i(s)P(s)}$$

3.2.2 Step 2: Outer-Loop Position Controller, G_o

Once the inner loop is stabalised, the effective outer-loop nominal plant is determined as:

$$P_o(s) = T_i(s)\frac{K}{s}$$

where $T_i(s)$ is the closed-loop speed transfer function and $\frac{K}{s}$ represents the plant's natural integrator that translates velocity into position.

The QFT method was again utilised to perform loop shaping for G_o to ensure that specification S2 and requirement R3 are met . Final verification was conducted using the overall position transfer function:

$$T_o(s) = \frac{G_o(s)P_o(s)}{1 + G_o(s)P_o(s)}$$

3.2.3 Step 3: Validation

To validate the controller, the Light Disk configuration was utilized to confirm whether the performance specifications were satisfied. The LD plant was selected because it represents the most sensitive system dynamics. Due to its minimal rotational inertia, the LD has the lowest physical resistance to changes in motion, making it the most susceptible to high-frequency oscillations and instability.

A sampling time of 40 ms was selected, falling within the recommended design range of 30 ms to 100 ms. This value was chosen to provide sufficient temporal resolution for the digital controller while addressing phase lag constraints identified in specification R3 in Table 3.1.

3.3 First Design Iteration

A comprehensive, step-by-step documentation of the design approach taken for this controller iteration is provided in the Cascade Design Report which was compliled by Group 20 [3].

Table 3.3 summarises the designed parameters and transfer functions used in this first iteration.

Table 3.3: Summary of the First Iteration Nominal Plants and Derived Controllers

Parameter	**Transfer Function**
Inner-loop nominal plant	$P(s) = \frac{47327}{0.1904s+1}$
Inner-loop digital controller (G_i)	$G_i(z) = 60$
Outer-loop digital controller (G_o)	$G_o(z) = \frac{1.752z-1.199}{z-0.4465}$

3.3.1 First Design Iteration Validation

The first iteration of the cascade controller design was characterized by several fundamental modelling errors that resulted in an unstable and impractical control system as illustrated in Figure 3.1.

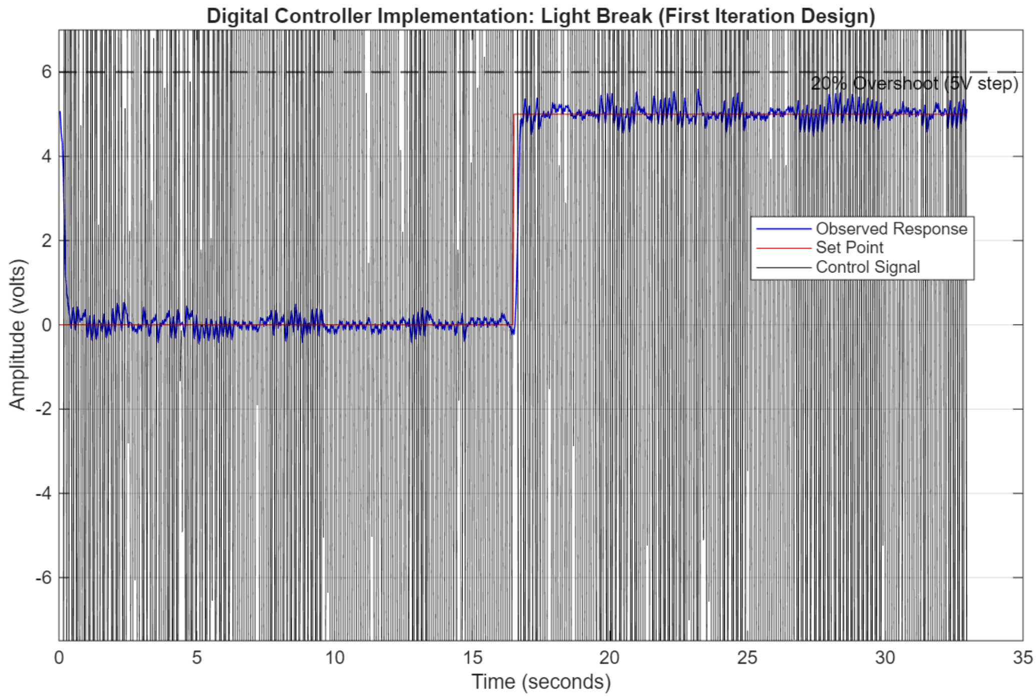


Figure 3.1: Iterative Tuning of G_i

The failure of the design primarily stemmed from the selection of an unrepresentative 'worst-case' nominal plant and an incorrect simulation structure that omitted physical hardware constraints. These modeling errors meant the simulation was unable to detect that the excessively high inner-loop gain of $G_i = 60$ would lead to constant control signal which is an inefficient method of control as it draws too much current. Additionally as it can be seen by the Observed Response graph, the system never reached a true settling point. A comprehensive audit of all the poor design decisions, including a color-coded analysis of the block diagram used for simulation and the root causes of the instability, is detailed in Appendix A.1.

This experience provided a critical learning opportunity, highlighting that the mastery of theoretical controller design steps must be matched by a deep understanding of the plant subsystems and their physical relationship to one another. Understanding how sensor constants translate physical quantities into voltages is essential for moving from an idealized model to a functional implementation.

3.4 Second Design Iteration

A critical error in first iteration analysis was the assumption that plant gain (A) and time constant (T) were independent variables by attempting to design for a synthetic 'worst-case' nominal plant. This error has been rectified by selecting the Heavy Disk plant.

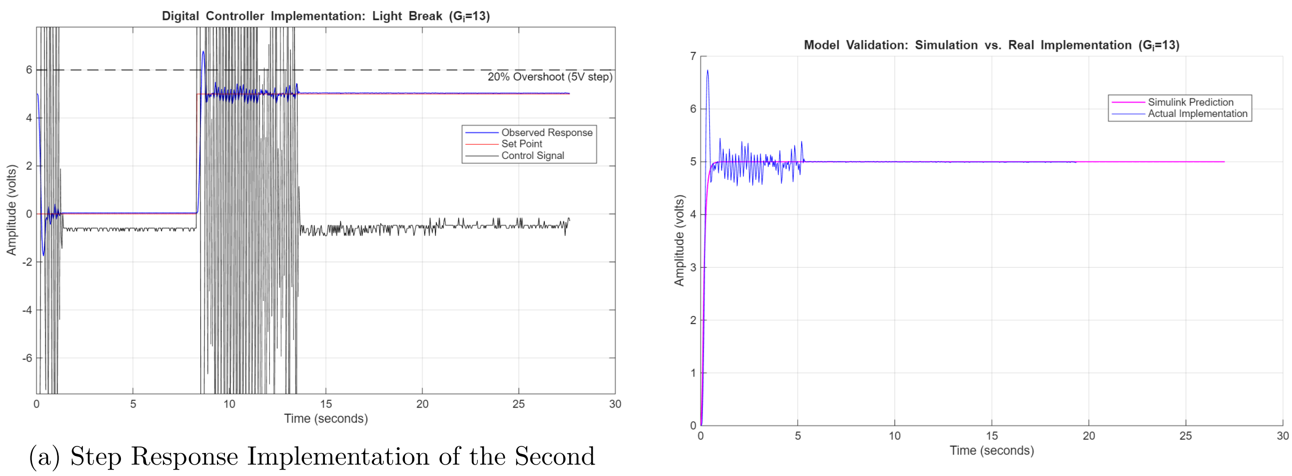
The Heavy Disk was selected because the mass off the disk introduces the largest rotational inertia and consequently the highest time constant. Designing for this scenario ensures that the controller tuned for the most difficult stability scenario.

Table 3.4: Summary of the Second Iteration Nominal Plants and Derived Controllers

Parameter	**Transfer Function**
Inner-loop nominal plant	$P(s) = \frac{47327}{2.1350s+1}$
Inner-loop digital controller (G_i)	$G_i(z) = 13$
Outer-loop digital controller (G_o)	$G_o(z) = \frac{2.394z-1}{z-0.4941}$

3.4.1 Second Design Iteration Validation

Figure 3.2 below illustrates the performance of the Second Iteration controller.



(a) Step Response Implementation of the Second Iteration Controller

(b) Simulink Simulation vs Real Implementation

Figure 3.2: Implementation of the Second Iteration Controller

As shown in Figure 3.2a, the inner-loop gain of $G_i = 13$ proved to be excessively aggressive, leading to severe actuator saturation and noise amplification. This is characterised by the black control signal, where the controller rapidly toggles between the ±10V DAC limits for approximately 5 seconds before the plant settles, thus characterising the controller as a 'bang-bang' controller. Although the response time was technically fast, the lack of damping and the resulting instability required a significant reduction in the inner-loop gain.

The discrepancy between the predicted and actual performance is further highlighted in Figure 3.2b. In the Simulink environment, the step test passed all criteria, producing a smooth, idealised response. However, this simulation environment represents an ideal design that fails to account for real-world uncertainties such as sensor noise, DC motor nonlinearities, such as saturation, as well as its mechanical constraints like rotational inertia. While the simulated control signal was mathematically accurate, the resulting control signal in implementation was physically non-realisable.

3.5 Final Design Iteration

Although the second iteration controller was deemed non-feasible due to the overshoot violation and the sustained control signal oscillations before settling, the underlying control structure proved effective because the tracking and settling time specifications were successfully met. This suggested that a complete redesign of the controller was unnecessary.instead, the design required a refinement of G_i.

While reducing the gain inherently results in a slightly slower rise time, the exceptional speed of the second design iteration provided a sufficient margin to allow for this trade-off.

Figure 3.3 illustrates the iterative tuning process conducted through trial and error on the physical servo plant.

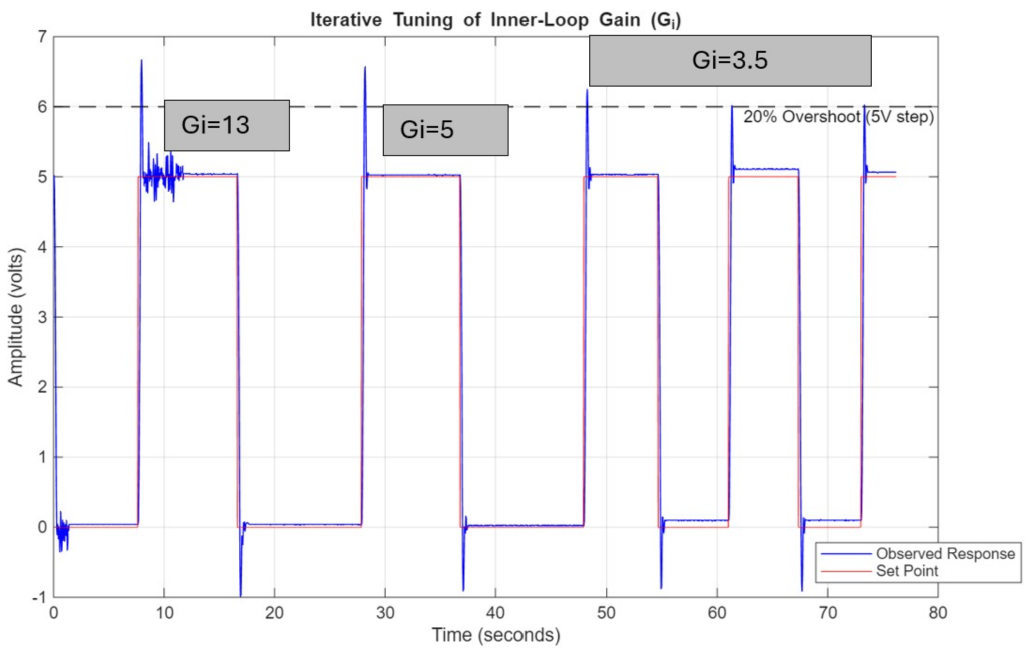
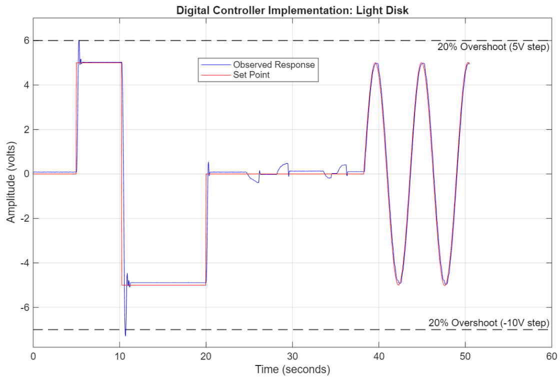


Figure 3.3: Iterative Tuning of G_i

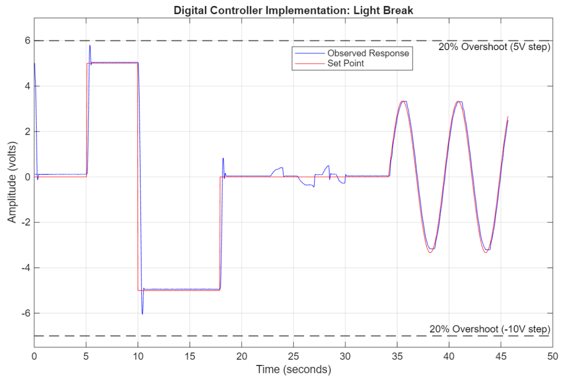
The final inner-loop gain was selected by tuning the hardware directly because it was established in Figure s3.2b that since a inner-loop gain of 13 misleadingly working in simulation as opposed to in real implementation, this would mean that testing in simulation was not an option.

Figure 3.3 illustrates that after decreasing G_i to 5, it was still infeasible because the overshoot specification was still violated, however, it confirmed that decreasing the gain was the correct strategy because settling oscillations were eliminated. A final inner-loop gain of $G_i = 3.5$ was selected as it provides a robust balance, ensuring that tracking, settling time, and overshoot specifications are met.

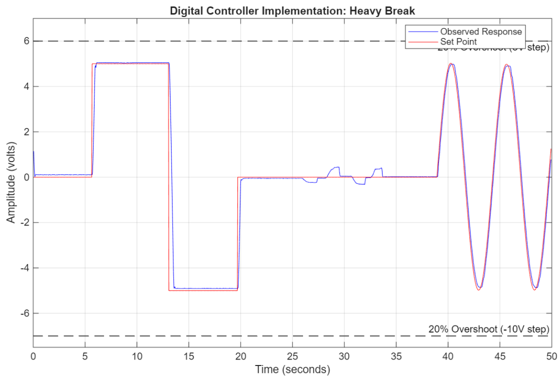
3.5.1 Final Design Iteration Validation



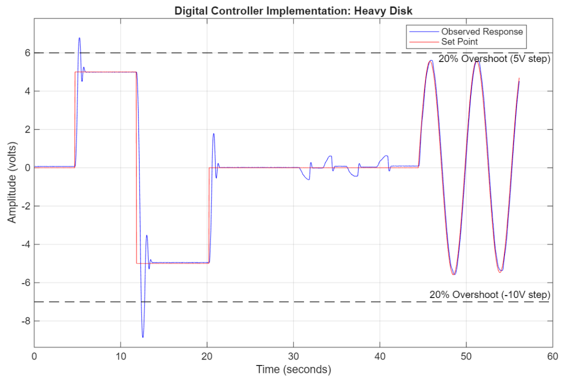
(a) Light Disk (LD)



(b) Light Brake (LB)



(c) Heavy Brake (HB)



(d) Heavy Disk (HD)

Figure 3.4: Simulink Step Responses for Cascaded Controller Model

As seen in Figure 3.4, the controller effectively tracks the setpoints across all plant configurations for both step and sine wave inputs. This confirms the robustness of the design, ensuring that specification S2 is satisfied across the entire uncertainty template. Furthermore, disturbance rejection tests conducted post-step indicate that the controller possesses high-speed rejection capabilities for all plant cases, successfully meeting specification S1.

Additionally, Time domain specifications R5 and R6 have been met as well.

Heavy Disk Overshoot Violation

According to Figure 3.4d, there is an overshoot violation for both the 5V step and -10V step. This violation is an indication that there the controller is constrained such that it is unable to meet the HD 20% overshoot specifications whilst simultaneously meeting all other specifications for each plant configureation. The reason for this blatant violation is the fact that the HD has the highest rotational inertia becasude the momentum of its heavy mass keeps it spinning past the target.

Heavy Disk Overshoot Analysis

According to Figure 3.4d, an overshoot violation is observed for both the 5V and −10V step inputs. This violation indicates that the controller is physically constrained as it is infeasible to meet the 20% overshoot limit for the Heavy Disk while simultaneously satisfying the performance requirements for the other three plant configurations.The primary cause for this violation is the Heavy Disk's high rotational inertia. The momentum generated by its large mass forces the disk to spin past the target setpoint, even as the controller attempts to decelerate the motor. In contrast, the Heavy Break configuration utilises high friction to physically 'assist' the controller in stopping the disk, leading to a much cleaner response in that specific case.

Rectification Discussion & Trade-offs

To strictly satisfy the overshoot constraint for the HD configuration, the inner-loop gain, G_i, would need to be decreased further. However, this presents two unacceptable engineering trade-offs:

- **Reduced Responsiveness:** A lower gain would significantly slow the response time, causing the controller to struggle particularly in high-friction scenarios, such as the Heavy Brake (HB) plant.
- **Tracking Errors:** Lowering the gain further would result in significant steady-state tracking errors across the entire template, compromising the system's primary functional requirements.

An alternative solution, which is outside the scope of the current hardware integration, would be to integrate an external braking mechanism to assist with damping the system without adjusting the inner-loop gain was considered. It would work by detecting a significant inertial disturbance and oscillation in the control signal to detect a struggle for the controller to control a heavy disk mass of the motor.

Ultimately, the chosen solution was to prioritise global template stability over the local transient perfection of the HD configuration as the client required that the LD plant specifications are have the highest priority.

Chapter 4

Conclusion

4.1 Conclusion

The implementation of the Final Design Iteration digital cascade controller successfully demonstrated a robust solution to regulate the speed and position of the DC servo motor.

Key insights from the design and validation process include the following:

- **Model Accuracy:** Fundamental errors made in the completion of the First Design Iteration emphasised the importance of dilligence when designing a model and simulation environment for a system, as the failure of that iteration showed the integral role and accurate model plays when designing a controller.
- **Simulation vs Reality Gap:** Discrepancies observed between the idealised Simulink environment and the physical implementation highlighted the critical impact of hardware constraints. This emphasised the need to determine a range of acceptible control parameters to ensure that the controller is robust and ready for any discrepencies and the need to account for nonlinearities and mechanical constraints of a system.
- **Constraint Hierarchy:** By prioritising the (LD) specifications as per the client's requirements, the final controller was optimised for the most sensitive plant case. The minor overshoot violations in the HD configuration were accepted as a trade-off for maintaining responsiveness in the the other three configuration.

It is recommended that Model Predictive Control (MPC) methodologies such as Dynamic Matrix Control(DMC) and Steady-State Optimisation are implemented to aid in improving the overall performance of the controller.

The resulting cascade-controlled closed-loop model of the DC servo system in Appendix B provides a stable foundation for the final phase of development, where the voltage-based control signals will be integrated into a functional Human-Machine Interface by utilising the sensor constants determined in Section 2.2.

Appendix A

Robust Controller Design

A.1 First Iteration: Poor Design Decisions

Table A.1: Analysis of Failure: First Iteration Structural and Modeling Flaws

Issue ID	Identified Issue	Root Cause	Observed Effect
Yellow	Extremely high inner loop gain ($G_i = 60$).	Selection of the 'worst-case' plant configuration as the nominal model.	Creation of a controller that constantly saturates the DAC operational limits of ± 10V.
Pink	Incorrect nominal plant model.	Assumption that gain (A) and time constant (T) are independent variables, ignoring physical motor dynamics.	Poor design of the cascade architecture due to unrepresentative plant modeling.
Orange	Incorrect consideration of sensor constants in the design.	Lack of understanding that controllers regulate voltage quantities while sensors translate physical measurements into voltages.	Flawed feedback logic where the signal feeding into $G_o(z)$ was an invalid representation of error.
Green	Omission of Saturation and Zero-Order Hold (ZOH) blocks.	Failure to model the physical constraints and digital sampling behavior of the hardware.	Flawed simulation that failed to predict real-world instability or actuator saturation.
Purple	Generation of a 'Bang-Bang' controller.	Failure to place scopes on critical internal signals like $u[k]$ to monitor actuator effort.	Inefficient and potentially damaging method of controlling the servo motor.

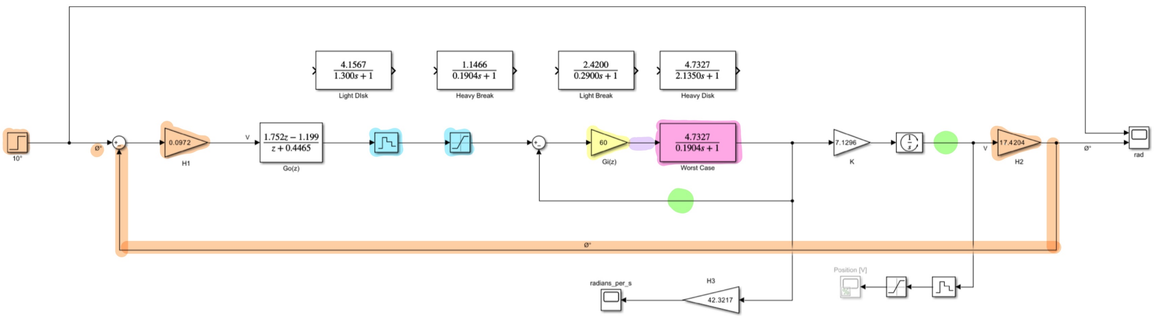
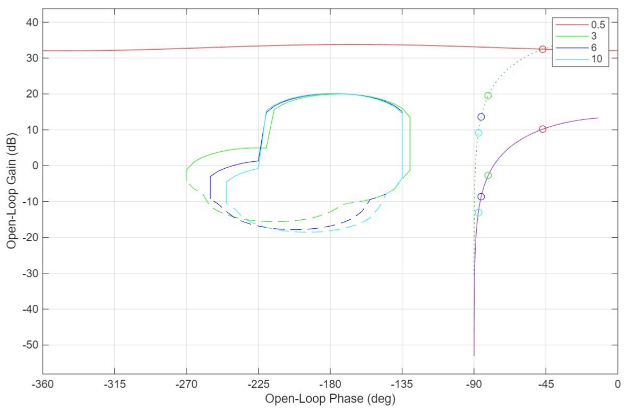


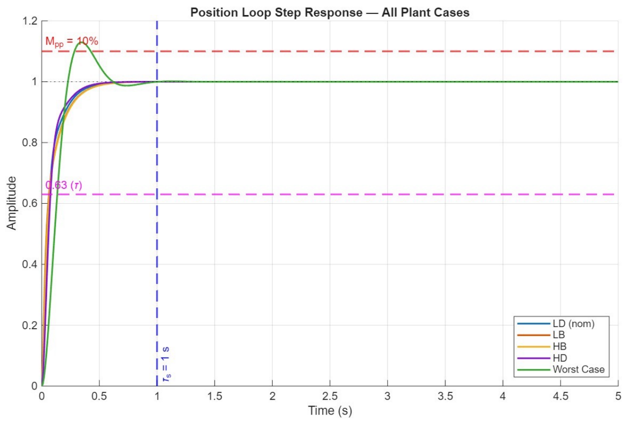
Figure A.1: First Iteration Simulink Model

A.2 Second iteration

A.2.1 Inner-Loop Controller Design



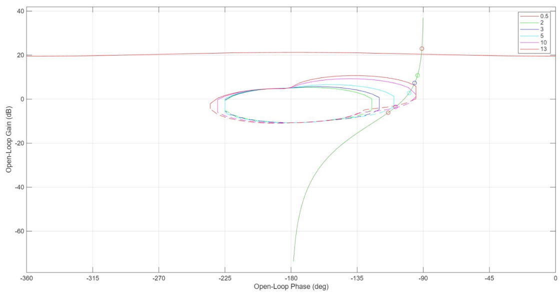
(a) Inverse Nichols Chart of the Inner-Loop Controller Design



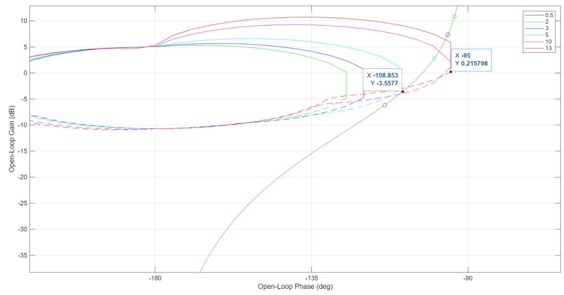
(b) Simulink Inner-Loop Step Response Test for all plant cases

Figure A.2: Simulink Inner-Loop Controller Design and Validation

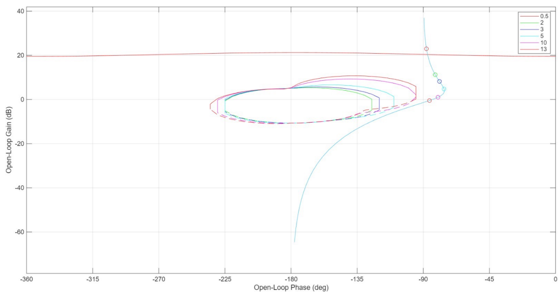
A.2.2 Outer-Loop Controller Design



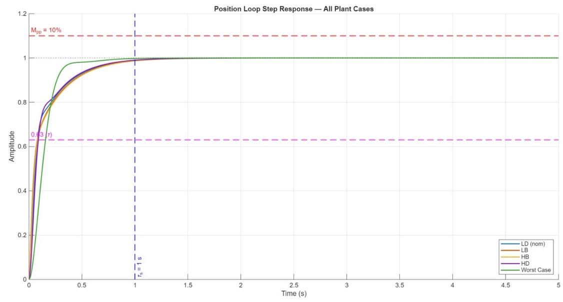
(a) Inverse Nichols Chart of the outer position loop with no controller



(b) Close-up of the crossover region used to determine the required phase lead



(c) Inverse Nichols Chart of the outer position loop with lead compensator controller



(d) Simulink Out-Loop Step Response Test for all plant cases

Figure A.3: Simulink Outer-Loop Controller Design and Validation

Appendix B

Final Cascade-Controlled Closed-Loop Model

B.1 Simulink Model

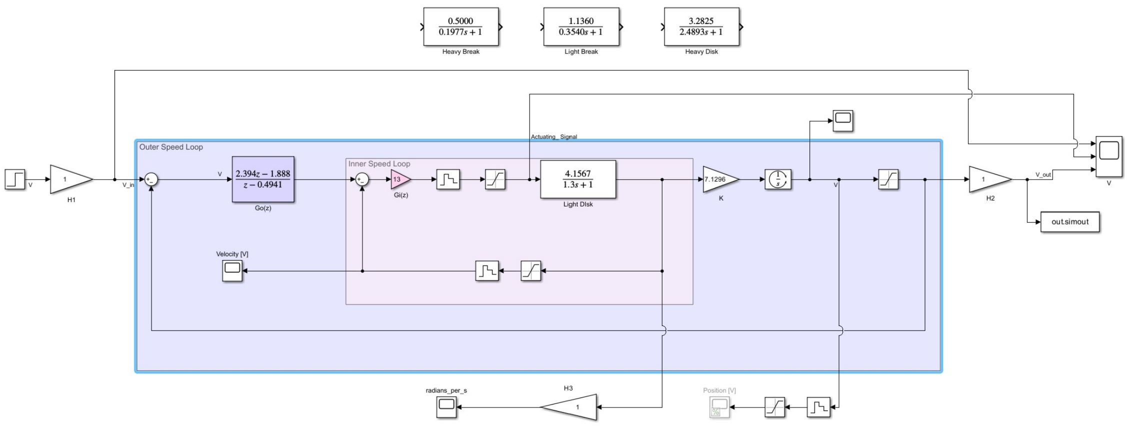


Figure B.1: Final Simulink Model

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[1] *EEE3094S/EEE3099S Mechatronics Project Manual: Servo Motor Control and Implementation*, University of Cape Town, Department of Electrical Engineering, 2026, course Reference Material.

[2] Group 20, "System identification report," University of Cape Town, Technical Report, 2026.

[3] ——, "Cascade control design report," University of Cape Town, Technical Report, 2026.